



Basic Electrical Quantities

V · I · R · P · E · f · C · L

• ELECTRICAL ENGINEERING • LESSON 02

The Language of Electricity — Six Core Quantities

Before you can design, install, or troubleshoot any electrical system you must speak the language of electricity fluently. That language has six words: **Voltage (V)**, **Current (I)**, **Resistance (R)**, **Power (P)**, **Energy (E)**, and **Frequency (f)**. Every electrical calculation, every equipment rating, and every safety rule is built from these six quantities.

Electric Charge and Current

Electric charge (Q) is the fundamental property of matter that causes it to experience electromagnetic force. The carrier of charge in a conductor is the **electron**, which carries a charge of -1.6×10^{-19} coulombs. One coulomb (C) = 6.24×10^{18} electrons.

Current (I) is the rate of flow of electric charge past a point. 1 ampere = 1 coulomb per second. In metals, electrons flow from negative to positive terminal (electron flow). By convention we use **conventional current** which flows from positive to negative — the direction a positive charge would move.

Voltage and the Water Analogy

Voltage (V), also called electromotive force (EMF) or potential difference, is the energy per unit charge. It is the "pressure" that pushes electrons through a conductor. Without a voltage difference, no current flows — just as water does not flow without a height difference or pump pressure.

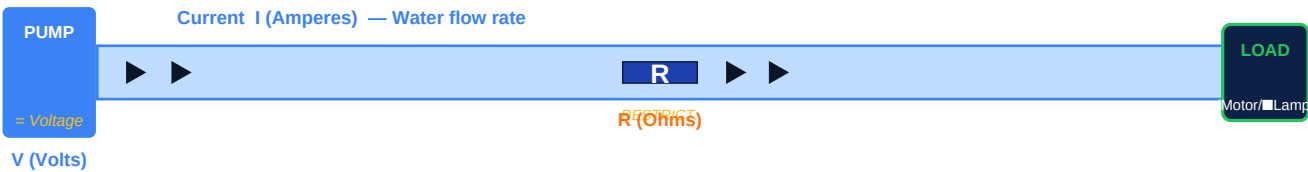


Figure 1 — Water Pipe Analogy: V = Pressure, I = Flow, R = Pipe Restriction

Quantity	Symbo l	Unit	Formula	Practical Example
Electric Charge	Q	Coulomb (C)	$Q = I \times t$	5 A flowing for 2 s delivers 10 C of charge
Current	I	Ampere (A)	$I = Q / t = V / R$	A 60 W lamp at 230 V draws 0.26 A
Voltage	V	Volt (V)	$V = I \times R = P / I$	UK mains = 230 V; car battery = 12 V; 9 V battery
Resistance	R	Ohm (Ω)	$R = V / I = V^2 / P$	A 100 W, 230 V lamp has $R = 529 \Omega$ when hot
Power	P	Watt (W)	$P = V \times I = I^2 R = V^2 / R$	2 kW kettle = 2 000 W; 100 W lamp = 100 W
Energy	E	Joule (J) or kWh	$E = P \times t$ (1 kWh = 3.6 MJ)	2 kW kettle for 3 min uses 360 000 J = 0.1 kWh

Memory trick for Ohm's Law — the VIR triangle: Cover the quantity you want to find. What remains is the formula. Cover V → $I \times R$. Cover I → V / R . Cover R → V / I .

Power, Energy and the Ohm's Law Wheel

Power (P) is the rate at which electrical energy is converted to another form (heat, light, motion). It is measured in watts (W). **Energy (E)** is power multiplied by time — the total work done. Your electricity bill charges you for energy in kilowatt-hours (kWh), not power in watts.

The four quantities V, I, R, and P are mathematically linked. If you know any two, you can calculate the other two. The formula wheel below shows all twelve possible relationships — memorise the four core formulas ($V=IR$, $P=VI$, $P=I^2R$, $P=V^2/R$) and the rest follow automatically.

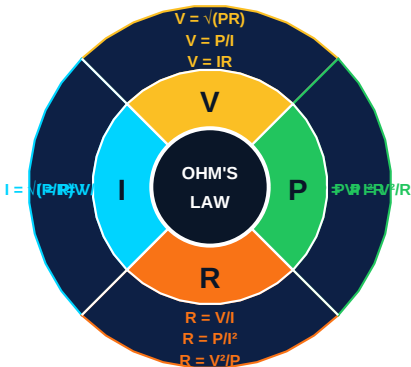


Figure 2 — Ohm's Law & Power Formula Wheel (V, I, R, P)

Worked Examples

Problem	Given	Find	Solution
Lamp circuit	V = 230 V, R = 529 Ω	I and P	$I = V/R = 230/529 = 0.435\text{ A}$ $P = VI = 230 \times 0.435 = 100\text{ W}$
Electric heater	P = 3 000 W, V = 230 V	I and R	$I = P/V = 3000/230 = 13.04\text{ A}$ $R = V/I = 230/13.04 = 17.6\text{ }\Omega$
Motor cable loss	I = 20 A, cable R = 0.5 Ω	Voltage drop and Power loss	$V_{\text{drop}} = IR = 20 \times 0.5 = 10\text{ V}$ $P_{\text{loss}} = I^2R = 400 \times 0.5 = 200\text{ W}$
Energy cost	P = 2 000 W (kettle), t = 5 min, cost = 30 p/kWh	Energy used and cost	$E = 2\text{ kW} \times (5/60)\text{ h} = 0.167\text{ kWh}$ Cost = $0.167 \times 30 = 5.0\text{ p}$

Resistance — What Causes It and How to Calculate It

Factor	Effect on Resistance	Formula Link
Material (resistivity ρ)	Copper: $\rho = 1.7 \times 10^{-8}\text{ }\Omega\cdot\text{m}$ (low — good conductor) Steel: $\rho = 1.0 \times 10^{-7}\text{ }\Omega\cdot\text{m}$ (6× higher)	$R = \rho L/A$
Length (L)	Longer wire = higher resistance. Double the length → double the resistance	$R \propto L$
Cross-section Area (A)	Thicker wire = lower resistance. Double the area → half the resistance	$R \propto 1/A$
Temperature	Most metals: resistance increases with temperature (positive temperature coefficient)	$R_T = R_{20}(1 + \alpha\Delta T)$

Cable sizing example: A 2.5 mm² copper cable, 20 m long. $\rho = 1.72 \times 10^{-8}\text{ }\Omega\cdot\text{m}$, $A = 2.5 \times 10^{-6}\text{ m}^2$. $R = \rho L/A = (1.72 \times 10^{-8} \times 20) / 2.5 \times 10^{-6} = 0.138\text{ }\Omega$ per conductor (0.275 Ω round trip). At 20 A: voltage drop = 5.5 V.

AC Quantities — Frequency, Peak and RMS

In direct current (DC) systems, voltage and current are constant. In alternating current (AC) systems — used for virtually all mains power distribution — voltage and current vary sinusoidally with time. This introduces three new quantities: **frequency**, **peak value**, and **RMS value**.

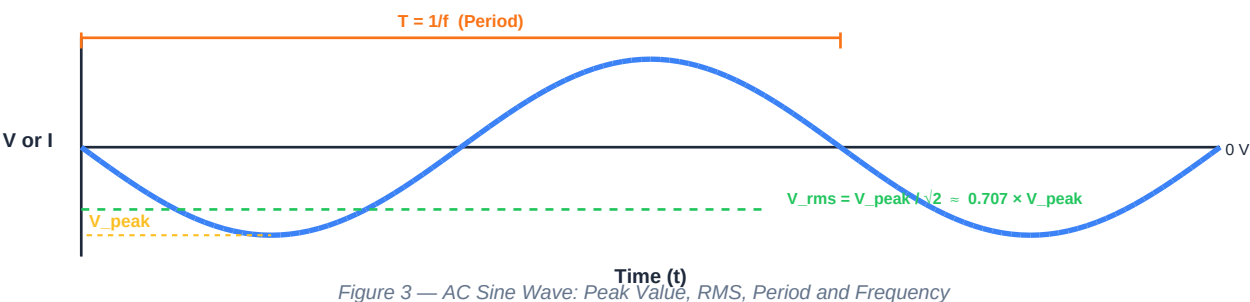


Figure 3 — AC Sine Wave: Peak Value, RMS, Period and Frequency

AC Quantity	Symbol	Definition	UK / Ethiopia Value
Frequency	f	Number of complete cycles per second. Unit: Hertz (Hz)	50 Hz (UK, EU, Africa, Middle East)
Period	T	Time for one complete cycle. $T = 1/f$	$T = 1/50 = 0.02 \text{ s} = 20 \text{ ms}$
Peak Voltage	V_{peak}	Maximum voltage reached in the cycle	$230 V_{rms} \times \sqrt{2} = 325 \text{ V peak}$ (UK mains)
Peak-to-Peak	V_{pp}	Voltage from positive to negative peak	$325 \times 2 = 650 \text{ V peak-to-peak}$
RMS Voltage	V_{rms}	Root Mean Square — DC equivalent that delivers same power $V_{rms} = V_{peak} / \sqrt{2} = 0.707 \times V_{peak}$	230 V_{rms} (UK mains nominal)
RMS Current	I_{rms}	Effective current value used in all power calculations $I_{rms} = I_{peak} / \sqrt{2}$	Used in all equipment ratings and cable sizing
Angular Frequency	ω	$\omega = 2\pi f$ — used in AC circuit equations	$\omega = 2\pi \times 50 = 314.2 \text{ rad/s}$

Capacitance and Inductance — Storage of Energy

Quantity	Symb ol	Unit	What It Does	Practical Use
Capacitance	C	Farad (F)	Stores energy in an electric field between two plates. Charges when voltage is applied; opposes changes in voltage.	Power factor correction, AC filters, motor start capacitors, energy storage
Inductive Reactance	X_L	Ohm (Ω)	Opposes change in current. $X_L = 2\pi fL$ — increases with frequency.	Transformers, motors, inductors in power supplies, chokes
Capacitive Reactance	X_C	Ohm (Ω)	Opposes change in voltage. $X_C = 1/(2\pi fC)$ — decreases with frequency.	AC coupling, filters, power factor correction banks
Impedance	Z	Ohm (Ω)	Total AC opposition: $Z = \sqrt{R^2 + (X_L - X_C)^2}$. Generalises resistance to AC circuits.	AC circuit analysis, motor impedance, transformer impedance

Master Reference — All Electrical Quantities

Symbo l	Quantity	SI Unit	Key Formula
Q	Electric Charge	Coulomb (C)	$Q = I \times t$

I	Current	Ampere (A)	$I = V/R = P/V$
V	Voltage / EMF	Volt (V)	$V = IR = P/I$
R	Resistance	Ohm (Ω)	$R = V/I = \rho L/A$
P	Power	Watt (W)	$P = VI = I^2R = V^2/R$
E	Energy	Joule (J) / kWh	$E = P \times t$
f	Frequency	Hertz (Hz)	$f = 1/T$
C	Capacitance	Farad (F)	$X_C = 1/(2\pi fC)$
L	Inductance	Henry (H)	$X_L = 2\pi fL$
Z	Impedance	Ohm (Ω)	$Z = \sqrt{R^2 + (X_L - X_C)^2}$
η	Efficiency	Dimensionless (%)	$\eta = P_{out} / P_{in} \times 100\%$
PF	Power Factor	Dimensionless (0–1)	$PF = \cos \phi = P / S$

What comes next — ELEC-003: Ohm's Law in depth — series and parallel circuits, Kirchhoff's voltage and current laws (KVL & KCL), and solving real circuit problems step by step.